

Лабораторная работа №1

Имитационное моделирование

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Раздел 1

Информация

- Ибрахим Мохсейн Алькамаль
- Российский университет дружбы народов

Раздел 2

Цель работы

- Подготовить рабочее окружение для имитационного моделирования.
- Освоить организацию проекта на языке Julia.
- Реализовать модель экспоненциального роста.
- Провести вычислительные эксперименты для различных значений параметра роста.
- Проанализировать полученные результаты.

Раздел 3

Настройка Git и средств аутентификации

Базовая настройка Git

- Проверена установка Git.
- Установлен GitHub CLI.
- Настроены глобальные параметры Git.

```
alkamal@fedora:~$ sudo dnf -y install git
[sudo] password for alkamal:
Updating and loading repositories:
Repositories loaded.
Package "git-2.55.0-1.fc44.x86_64" is already installed.

Nothing to do.
alkamal@fedora:~$ sudo dnf -y install gh
Updating and loading repositories:
Repositories loaded.
Package                               Arch      Version
Installing:
  gh                               x86_64    0:2.97.0-2.fc44

Transaction Summary:
  Installing:      1 package

Total size of inbound packages is 12 MiB. Need to download 12 MiB.
After this operation, 40 MiB extra will be used (install 40 MiB, remove 0 B).
[1/1] gh-0:2.97.0-2.fc44.x86_64
-----
[1/1] Total
Running transaction
[1/3] Verify package files
[2/3] Prepare transaction
```


Аутентификация на GitHub

- Выполнена аутентификация через GitHub CLI.
- Настроен протокол HTTPS для операций Git.

```
Press Enter to open https://github.com/login/device in your browser...  
✓ Authentication complete.  
- gh config set -h github.com git_protocol https  
✓ Configured git protocol  
✓ Logged in as Ebrahimalkamal2027  
alkamal@fedora:~$
```

Рисунок 2: Успешная аутентификация на GitHub через GitHub CLI

Создание SSH-ключа

- Проверено состояние авторизации командой `gh auth status`.
- Создана пара SSH-ключей Ed25519.

```
alkamal@fedora:~$ gh auth status
github.com
✓ Logged in to github.com account Ebrahimalkamal2027 (keyring)
- Active account: true
- Git operations protocol: https
- Token: gho_*****
- Token scopes: 'gist', 'read:org', 'repo', 'workflow'
alkamal@fedora:~$ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519-git -C "1032225432@rudn.ru"
Generating public/private ed25519 key pair.
Created directory '/home/alkamal/.ssh'.
Enter passphrase for "/home/alkamal/.ssh/id_ed25519-git" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/alkamal/.ssh/id_ed25519-git
Your public key has been saved in /home/alkamal/.ssh/id_ed25519-git.pub
The key fingerprint is:
SHA256:nM9dgmHM+84p74fUBsN0Arajac97t5yAEMTfxIBmJS0 1032225432@rudn.ru
The key's randomart image is:
+--[ED25519 256]--+
|      .o+oo      |
|      .E.+oo      |
|    o.+ ==.      |
|      o+oo+=      |
|      .oS.o+.+.   |
|      +. + oooo   |
|      . o. +.oo   |
```

Создание GPG-ключа

- Создан GPG-ключ с данными пользователя.
- Созданы открытый и секретный ключи.

```
Real name: alkamal
Email address: 1032225432@rudn.ru
Comment: key GPG
You selected this USER-ID:
    "alkamal (key GPG) <1032225432@rudn.ru>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? 0
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: /home/alkamal/.gnupg/trustdb.gpg: trustdb created
gpg: directory '/home/alkamal/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/alkamal/.gnupg/openpgp-revocs.d/9086B2906E6889B1394454B2932973874ABA5181.rev'
public and secret key created and signed.

pub   rsa4096 2026-09-01 [SC]
      9086B2906E6889B1394454B2932973874ABA5181
```

Рисунок 4: Создание GPG-ключа для подписания коммитов

Настройка GPG-подписания

- Проверен созданный GPG-ключ.
- Настроено автоматическое GPG-подписание коммитов.

```
alkamal@fedora:~$ gpg --list-secret-keys --keyid-format LONG
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
[keyboard]
-----
sec   rsa4096/932973874ABA5181 2026-09-01 [SC]
      9086B2906E6B89B1394454B2932973874ABA5181
uid           [ultimate] alkamal (key GPG) <1032225432@rudn.ru>
ssb   rsa4096/5508F0B4190FD931 2026-09-01 [E]

alkamal@fedora:~$ gpg --armor --export 932973874ABA5181 | wl-copy
alkamal@fedora:~$ gpg --armor --export 9086B2906E6B89B1394454B2932973874ABA5181 | wl-copy
alkamal@fedora:~$ git config --global user.signingkey 9086B2906E6B89B1394454B2932973874ABA5181
alkamal@fedora:~$ git config --global commit.gpgsign true
alkamal@fedora:~$ git config --global gpg.program $(which gpg)
alkamal@fedora:~$
```

Рисунок 5: Настройка GPG-подписания коммитов Git

Проверка инструментов

- Проверены установленные инструменты рабочего окружения.
- Подтверждено наличие commitizen, cz-customizable и standard-version.

```
alkamal@fedora:~$ quarto --version
rpm -q libxcrypt-compat
node --version
yarn --version
pnpm --version
git flow version
pnpm list -g --depth=0 | grep -E 'commitizen|cz-customizable|standard-version'
1.10.18
libxcrypt-compat-4.5.2-3.fc44.x86_64
v22.23.1
1.22.22
10.33.0
1.12.3 (AVH Edition)
├─ commitizen@4.3.2
├─ cz-customizable@7.5.4
└─ standard-version@9.5.0
```

Рисунок 6: Проверка версий и установленных инструментов рабочего окружения

Раздел 4

Подготовка рабочего пространства лабораторной работы

Клонирование репозитория

- Репозиторий курса клонирован с GitVerse по SSH.
- Создана локальная копия проекта.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling$ git clone ssh://git@gitverse.ru:2222/alkamal/2026-1--study--simulation-modeling.git
Cloning into '2026-1--study--simulation-modeling'...
remote: Enumerating objects: 41, done.
remote: Counting objects: 100% (41/41), done.
remote: Compressing objects: 100% (40/40), done.
remote: Total 41 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (41/41), 25.31 KiB | 1.05 MiB/s, done.
Resolving deltas: 100% (1/1), done.
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling$
```

Рисунок 7: Клонирование репозитория курса с GitVerse

Подготовка структуры проекта

- Выполнена команда `make prepare`.
- Загружены подмодули шаблонов презентации и отчёта.
- Проверена структура лабораторных работ.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ make prepare
Submodule 'template/presentation' (https://gitverse.ru/dharma/academic-presentation-markdown-template.git) registered for path 'template/presentation'
Submodule 'template/report' (https://gitverse.ru/dharma/academic-laboratory-report-template.git) registered for path 'template/report'
Cloning into '/home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/template/presentation'...
Cloning into '/home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/template/report'...
Submodule path 'template/presentation': checked out '60a6091ebddb01a8d98f2c5fae0f98dc63ad031'
Submodule path 'template/report': checked out '788b8c51acd94d4b78e7634198f221e5decb6355'
Synchronizing submodule url for 'template/presentation'
Synchronizing submodule url for 'template/report'
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ ls
CODE_OF_CONDUCT.md CONTRIBUTING.md COURSE labs LICENSE Makefile package.json prepare README.en.md README.git-flow.md README.md SECURITY.md template
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ ls labs
lab01 lab02 lab03 lab04 lab05 lab06 lab07 lab08 README.md README.ru.md
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$
```

Рисунок 8: Подготовка проекта и проверка структуры лабораторных работ

Синхронизация с GitVerse

- Изменения отправлены в GitVerse.
- Ветка `master` синхронизирована с `origin/master`.
- Рабочее дерево не содержит изменений.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git push
Enumerating objects: 64, done.
Counting objects: 100% (64/64), done.
Delta compression using up to 6 threads
Compressing objects: 100% (47/47), done.
Writing objects: 100% (60/60), 702.58 KiB | 54.04 MiB/s, done.
Total 60 (delta 15), reused 0 (delta 0), pack-reused 0 (from 0)
remote: . Processing 1 references
remote: Processed 1 references in total
To ssh://gitverse.ru:2222/alkamal/2026-1--study--simulation-modeling.git
   adcc186..2495977  master -> master
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
```

Рисунок 9: Отправка изменений в GitVerse и проверка состояния репозитория

- Ветка `master` отправлена в репозиторий GitHub.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git push github master
Enumerating objects: 101, done.
Counting objects: 100% (101/101), done.
Delta compression using up to 6 threads
Compressing objects: 100% (86/86), done.
Writing objects: 100% (101/101), 727.52 KiB | 4.72 MiB/s, done.
Total 101 (delta 17), reused 40 (delta 1), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (17/17), done.
To github.com:Ebrahimalkamal2027/2026-1--study--simulation-modeling.git
 * [new branch]      master -> master
```

Рисунок 10: Отправка ветки `master` в удалённый репозиторий GitHub

Инициализация Git Flow

- Инициализирован Git Flow.
- Основная ветка — master.
- Ветка разработки — develop.
- Префикс тегов версий — v.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/.git/hooks]
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$
```

Рисунок 11: Инициализация и настройка Git Flow

Создание релиза

- Ветки и тег v1.0.0 отправлены в GitVerse и GitHub.
- Сформирован CHANGELOG.md.
- Создан релиз v1.0.0.

```
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 2.78 KiB | 2.78 MiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: You can create a PR using the link below:
remote:   https://gitverse.ru/alkamal/2026-1--study--simulation-modeling/compare?baseBranch=master&headBranch=develop
remote:
remote: .. Processing 2 references
remote: Processed 2 references in total
To ssh://gitverse.ru:2222/alkamal/2026-1--study--simulation-modeling.git
   2495977..dfd87fb  develop -> develop
   2495977..d96e692  master -> master
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 168 bytes | 168.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: .. Processing 1 references
remote: Processed 1 references in total
To ssh://gitverse.ru:2222/alkamal/2026-1--study--simulation-modeling.git
 * [new tag]         v1.0.0 -> v1.0.0
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git push github --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 6 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 2.78 KiB | 2.78 MiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Ebrahimalkamal2027/2026-1--study--simulation-modeling.git
   2495977..dfd87fb  develop -> develop
   2495977..d96e692  master -> master
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git push github --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 168 bytes | 168.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Ebrahimalkamal2027/2026-1--study--simulation-modeling.git
 * [new tag]         v1.0.0 -> v1.0.0
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ mkdir -p ../release
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ cp CHANGELOG.md ../release
```

Раздел 5

Создание и проверка проекта моделирования

Создание проекта Julia

- Создан каталог `project`.
- Проверена структура проекта.
- Созданы каталоги `data`, `notebooks`, `plots`, `scripts`, `src` и `test`.

```
[14a3606d] + MozillaCerts_jll v2025.5.20
[458c3e95] + OpenSSL_jll v3.5.1+0
[83775a58] + Zlib_jll v1.3.1+2
[8e850ede] + nghttp2_jll v1.64.0+1
[3f19e933] + p7zip_jll v17.5.0+2
Проект создан: project
Перейдите в директорию: cd project
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01$ ls
presentation project report setup_project.jl
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01$ ls project/
data Manifest.toml notebooks papers plots Project.toml README.md _research scripts src test
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01$
```

Рисунок 13: Создание и проверка структуры проекта Julia

Установка зависимостей

- Установлены зависимости Julia.
- Выполнена предварительная компиляция пакетов.

```
[b77e0e4c] + InteractiveUtils v1.11.0
[4a954fe1] + LazyArtifacts v1.11.0
[37e2e46d] + LinearAlgebra v1.12.0
[9abb9845] + Profile v1.11.0
[3fa0cd96] + REPL v1.11.0
[9e88b42a] + Serialization v1.11.0
[6462fe0b] + Sockets v1.11.0
[2f01184e] + SparseArrays v1.12.0
[8dfe0614] + Test v1.11.0
[e66e0078] + CompilerSupportLibraries_jll v1.3.0+1
[4536629a] + OpenBLAS_jll v0.3.29+0
[05823500] + OpenLibm_jll v0.8.7+0
[efcefd77] + PCRE2_jll v10.44.0+1
[bea87d4a] + SuiteSparse_jll v7.8.3+2
[8e850b96] + libblastrampoline_jll v5.15.0+0

Info Packages marked with X have new versions available but compatibility constraints restrict them from upgrading. To see why use `status --outdated -m`
Building Conda → ~/.julia/scratchespaces/44cfe95a-1eb2-52ea-b672-e2afdf69b78f/8f06b8cfa4c514c7b9546756dbae91fcfb9c92dc9/build.log
Building Julia → ~/.julia/scratchespaces/44cfe95a-1eb2-52ea-b672-e2afdf69b78f/102656c4efc9737f892e1bca7e66ae374c650746/build.log
Precompiling packages finished.
290 dependencies successfully precompiled in 288 seconds. 58 already precompiled.
2 dependencies had output during precompilation:
┌ LinearSolve
│   Downloading artifact: NKL
└─
┌ MKL_jll
│   Downloading artifact: IntelOpenMP
│   Downloading artifact: oneTBB
└─

Все пакеты установлены!
Для проверки: using DrWatson, DifferentialEquations, Plots
alkamal@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1--study==simulation-modeling/labs/lab01/project$
```

Рисунок 14: Установка и предварительная компиляция пакетов Julia

Проверка окружения

- Выполнен `scripts/test_setup.jl`.
- Проверена доступность необходимых пакетов.
- Проверена структура каталогов проекта.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ julia --project=. scripts/test_setup.jl
✔ Проект активирован: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project

Проверка пакетов:
✔ DrWatson
✔ DifferentialEquations
✔ Plots
✔ Dataframes
✔ CSV
✔ JLD2
✔ Literate
✔ IJulia
✔ BenchmarkTools
✔ Quarto

Структура проекта:
Корень: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project
Данные: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/data
Скрипты: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/src
Графики: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/plots
```

Рисунок 15: Проверка окружения и установленных пакетов проекта

Раздел 6

Реализация модели экспоненциального роста

Выполнение модели

- Создан сценарий 01_exponential_growth.jl.
- Получены первые значения решения модели.
- Рассчитано время удвоения: 2.31.

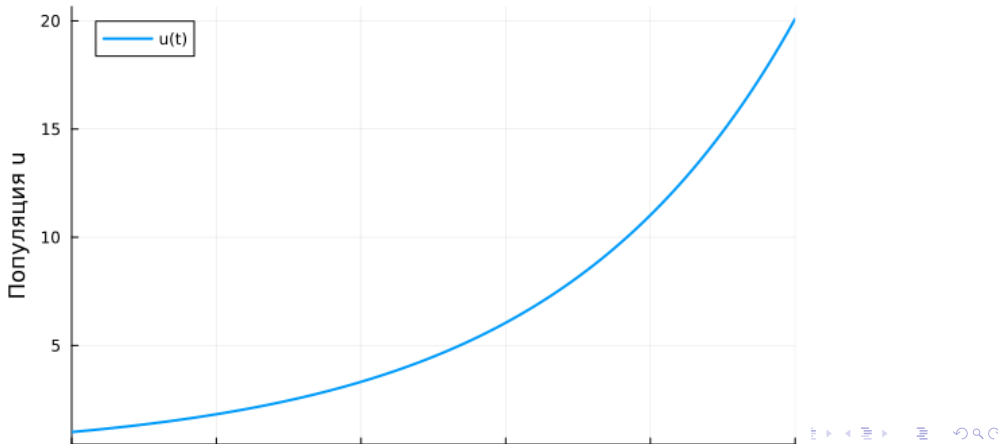
```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ cd scripts/
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts$ nano 01_exponential_growth.jl
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts$ cd ..
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ julia --project=. scripts/01_exponential_growth.jl
Could not create decoration from factory! Running with no decorations.
Первые 5 строк результатов:
5x2 DataFrame
  Row  | t      u
      |----|----
1     | 0.0    1.0
2     | 0.1    1.83845
3     | 0.2    1.86184
4     | 0.3    1.89417
5     | 0.4    1.1275
Аналитическое время удвоения: 2.31
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls plots/
'exponential_growth_a=0.3.png'
```

Рисунок 16: Выполнение модели экспоненциального роста

Визуализация модели

- Построен график для $\alpha = 0.3$.
- Получен экспоненциальный рост популяции во времени.

Экспоненциальный рост ($\alpha = 0.3$)



Сохранение результатов

- Результаты сохранены в `all_results.jld2`.
- График сохранён в каталоге результатов.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ nano scripts/01_exponential_growth.jl
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ julia --project=. scripts/01_exponential_growth.jl
Could not create decoration from factory! Running with no decorations.
Параметр 5 строк результатов:
5x2 DataFrame
  Row  t      u
  Float64  Float64
1      0.0  1.0
2      0.1  1.83845
3      0.2  1.86184
4      0.3  1.89417
5      0.4  1.1275

Аналитическое время удвоения: 2.31
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls plots/
01_exponential_growth  'exponential_growth_a=0.3.png'
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls data/01_exponential_growth/
all_results.jld2
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls plots/01_exponential_growth/
'exponential_growth_a=0.3.png'
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$
```

Рисунок 18: Проверка сохранённых данных и графика модели

Раздел 7

Литературная реализация модели

Генерация файлов

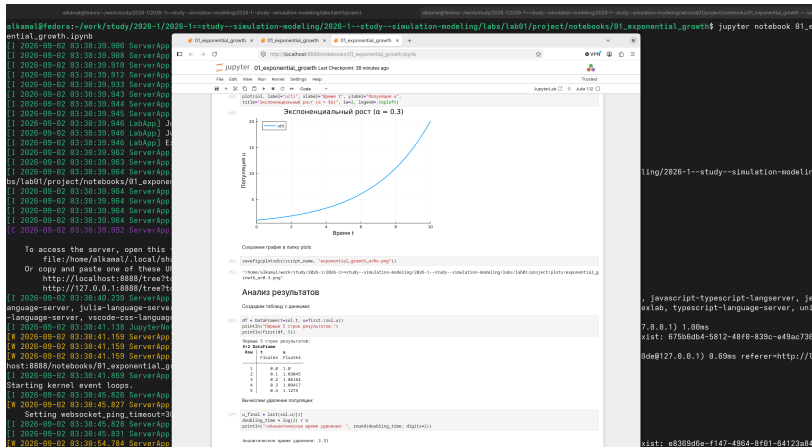
- Подготовлен сценарий `tangle.jl`.
- Сформирован чистый код Julia.
- Созданы документы `.qmd` и `.ipynb`.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ nano scripts/tangle.jl
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls scripts/
01_exponential_growth.jl  Intro.jl  tangle.jl  test_setup.jl
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ julia --project=. scripts/tangle.jl scripts/01_exponential_growth.jl
Генерация из: scripts/01_exponential_growth.jl
[ Info: generating plain script file from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/01_exponential_growth.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/01_exponential_growth/01_exponential_growth.jl"
✓ Чистый скрипт: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/01_exponential_growth/01_exponential_growth.jl
[ Info: generating markdown page from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/01_exponential_growth.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/markdown/01_exponential_growth/01_exponential_growth.qmd"
✓ Quarto: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/markdown/01_exponential_growth/01_exponential_growth.qmd
[ Info: generating notebook from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/01_exponential_growth.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/notebooks/01_exponential_growth/01_exponential_growth.ipynb"
✓ Notebook: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/notebooks/01_exponential_growth/01_exponential_growth.ipynb
Готово! Все файлы созданы.
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls notebooks/
01_exponential_growth
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls notebooks/01_exponential_growth/
01_exponential_growth.ipynb
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls markdown/01_exponential_growth/
01_exponential_growth.qmd
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$
```

Рисунок 19: Генерация кода, документации Quarto и Jupyter Notebook

Выполнение Jupyter Notebook

- Выполнен 01_exponential_growth.ipynb.
- Получены график и таблица результатов.
- Получено время удвоения 2.31.



Раздел 8

Проведение вычислительного эксперимента

Параметрическое сканирование

- Выполнен 02_exponential_growth.jl.
- Базовый эксперимент проведён при $\alpha = 0.3$.
- Исследованы $\alpha = 0.1, 0.3, 0.5, 0.8, 1.0$.
- Результаты сведены в таблицу.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ julia --project=. scripts/02_exponential_growth.jl
Базовые параметры эксперимента:
α = 0.3
u0 = [1.0]
saveat = 8.1
solver = Tsit5{typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!), FastBroadcast.Serial{OrdinaryDiffEqCore.trivial_limiter!, OrdinaryDiffEqCore.trivial_limiter!, FastBroadcast.Serial{}}}
experiment_name = base_experiment
tspan = (0.0, 10.0)
[ Info: File /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/data/02_exponential_growth/single/exp_growth_experiment_name=base_experiment_saveat=8.1_α=0.3.jld2 does not exist. Producing it now..
[ Info: File /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/data/02_exponential_growth/single/exp_growth_experiment_name=base_experiment_saveat=8.1_α=0.3.jld2 saved.

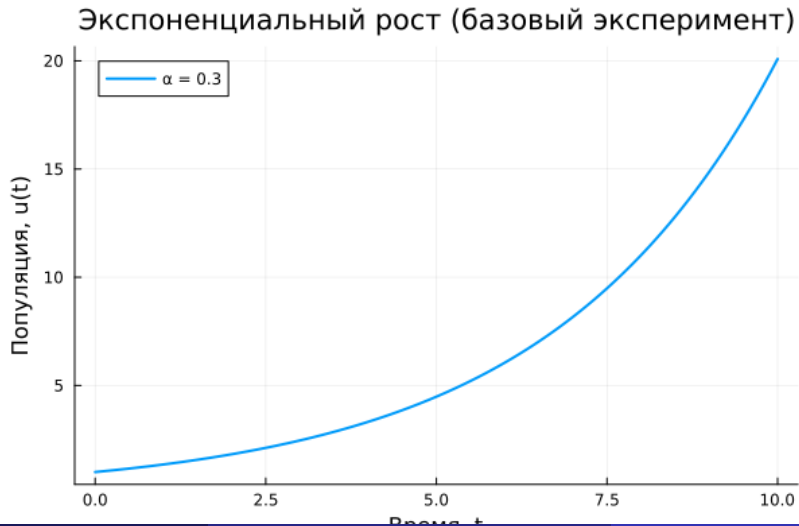
Результаты базового эксперимента:
Финальная популяция: 28.88544851618676
Время удвоения: 2.31
Файл результатов: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/data/02_exponential_growth/single/exp_growth_experiment_name=base_experiment_saveat=8.1_α=0.3.jld2
Could not create decoration from factory! Running with no decorations.

=====
ПАРАМЕТРИЧЕСКОЕ СКАНИРОВАНИЕ
Всего комбинаций параметров: 5
Исследуемые значения α: [0.1, 0.3, 0.5, 0.8, 1.0]
=====
Warning: Assignment to `data` in soft scope is ambiguous because a global variable by the same name exists: `data` will be treated as a new local. Disambiguate by using `local data` to suppress this warning or `global data` to assign to the existing global variable.
Warning: Assignment to `path` in soft scope is ambiguous because a global variable by the same name exists: `path` will be treated as a new local. Disambiguate by using `local path` to suppress this warning or `global path` to assign to the existing global variable.
julia> j -W ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/02_exponential_growth.jl:188
Progress: 1/5 | α = 0.1
Progress: 2/5 | α = 0.3
Progress: 3/5 | α = 0.5
Progress: 4/5 | α = 0.8
Progress: 5/5 | α = 1.0

Сводная таблица результатов:
5x3 DataFrame
Row | α          final_population  doubling_time
----|-----
1    | 0.1        1.0000000000000002  1.0000000000000002
2    | 0.3        28.88544851618676  2.3100000000000002
3    | 0.5        10.000000000000001  1.0000000000000002
4    | 0.8        3.0000000000000004  1.0000000000000002
5    | 1.0        2.0000000000000004  1.0000000000000002
```

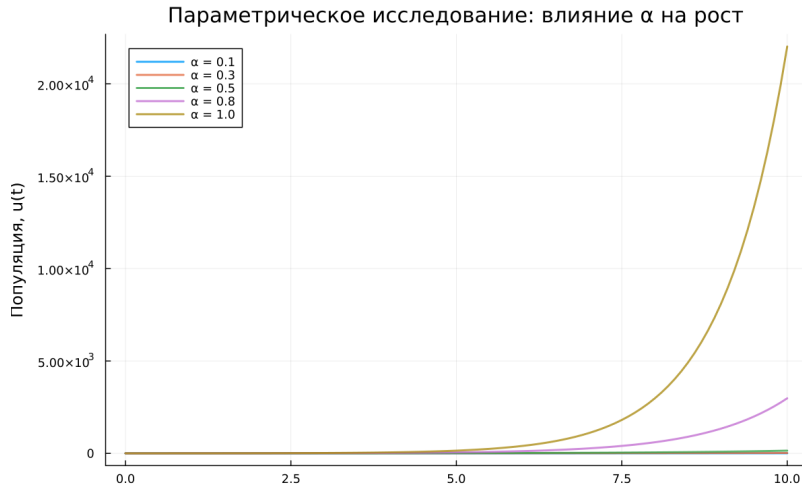
Базовый эксперимент

- Построен график базового эксперимента при $\alpha = 0.3$.



Сравнение параметров модели

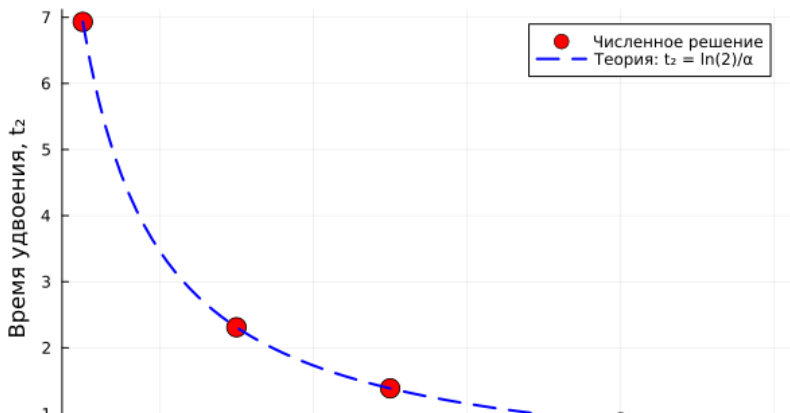
- Выполнено сравнение результатов для различных α .
- Увеличение α приводит к более быстрому росту популяции.



Анализ времени удвоения

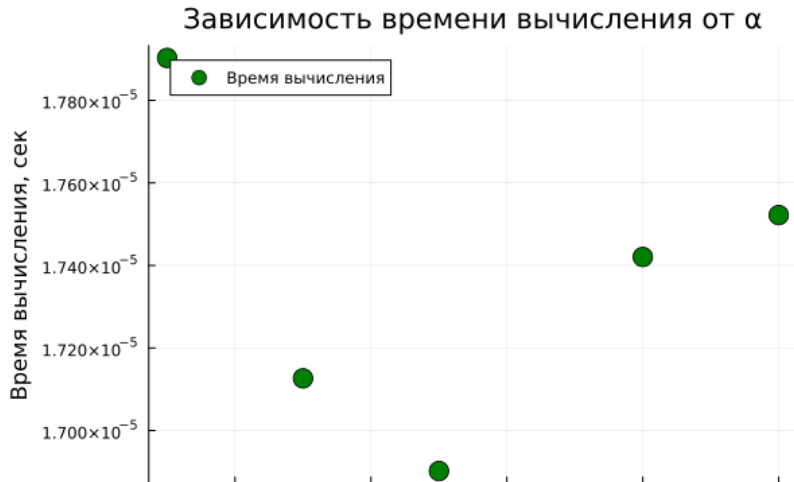
- Построена зависимость времени удвоения от α .
- Численные результаты сравнены с $t_2 = \ln(2)/\alpha$.
- При увеличении α время удвоения уменьшается.

Зависимость времени удвоения от α



Анализ времени вычисления

- Построена зависимость времени вычисления от α .
- Сравнено время вычислений для пяти значений параметра.



Раздел 9

Литературная реализация вычислительного эксперимента

Генерация файлов эксперимента

- 02_exponential_growth.jl преобразован с помощью tangle.jl.
- Создан чистый сценарий Julia.
- Созданы документ Quarto и Jupyter Notebook.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1--study==simulation-modeling/labs/lab01/project$ julia --project=. scripts/tangle.jl scripts/02_exponential_growth.jl
Генерация us: scripts/02_exponential_growth.jl
[ Info: generating plain script file from '~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1--study==simulation-modeling/labs/lab01/project/scripts/02_exponential_growth.jl'
[ Info: writing result to '~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1--study==simulation-modeling/labs/lab01/project/scripts/02_exponential_growth/02_exponential_growth.jl'
✓ Четкий скрипт: /home/alkamal/work/study/2026-1/2026-1==study==simulation-modeling/2026-1--study==simulation-modeling/labs/lab01/project/scripts/02_exponential_growth/02_exponential_growth.jl
[ Info: generating markdown page from '~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1--study==simulation-modeling/labs/lab01/project/scripts/02_exponential_growth.jl'
[ Info: writing result to '~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1--study==simulation-modeling/labs/lab01/project/markdown/02_exponential_growth/02_exponential_growth.qmd'
✓ Quarto: /home/alkamal/work/study/2026-1/2026-1==study==simulation-modeling/2026-1--study==simulation-modeling/labs/lab01/project/markdown/02_exponential_growth/02_exponential_growth.qmd
[ Info: generating notebook from '~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1--study==simulation-modeling/labs/lab01/project/scripts/02_exponential_growth.jl'
[ Info: writing result to '~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1--study==simulation-modeling/labs/lab01/project/notebooks/02_exponential_growth/02_exponential_growth.ipynb'
✓ Notebook: /home/alkamal/work/study/2026-1/2026-1==study==simulation-modeling/2026-1--study==simulation-modeling/labs/lab01/project/notebooks/02_exponential_growth/02_exponential_growth.ipynb
Готово! Все файлы созданы.
```

Рисунок 26: Генерация файлов литературной реализации вычислительного эксперимента

Выполнение параметрического исследования

- Выполнен 02_exponential_growth.ipynb.
- Получены результаты параметрического исследования.
- Выполнено сравнение численного и теоретического времени удвоения.



Раздел 10

Выводы

- Подготовлено рабочее окружение.
- Настроены Git, GitHub и GitVerse.
- Создан и проверен проект Julia.
- Реализована модель экспоненциального роста.
- Проведено параметрическое исследование.
- При увеличении λ рост популяции ускоряется.
- При увеличении λ время удвоения уменьшается.
- Сформированы сценарии, документация Quarto и Jupyter Notebook.